

Qatar-1b

R-1829

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_150510 · created 2026-08-03T05:38:53+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2457152.85987 +/- 0.00053 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.096 +/- 0.017
Transit depth (Rp/Rs)^2	0.92 +/- 0.33 [%]
Orbital Inclination (inc)	84.85 +/- 1.53
Airmass coefficient 1 (a1)	1.004 +/- 0.012
Airmass coefficient 2 (a2)	-0.0028 +/- 0.0068
Scatter in the residuals of the lightcurve fit is	1.24 %
Best Comparison Star	#3 - [311, 142]
Optimal Aperture	3.19
Optimal Annulus	7.34
Transit Duration (day)	0.0673 +/- 0.008

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.096 $\pm$ 0.017 vs 0.1463 (deviation 2.48 σ)
Duration fit vs published	0.0673 d vs 0.0692 d (deviation 0.2 σ)
Mid-time O-C	1.3 min
Depth SNR	4.7
Residual scatter	1.24 %

Light curve

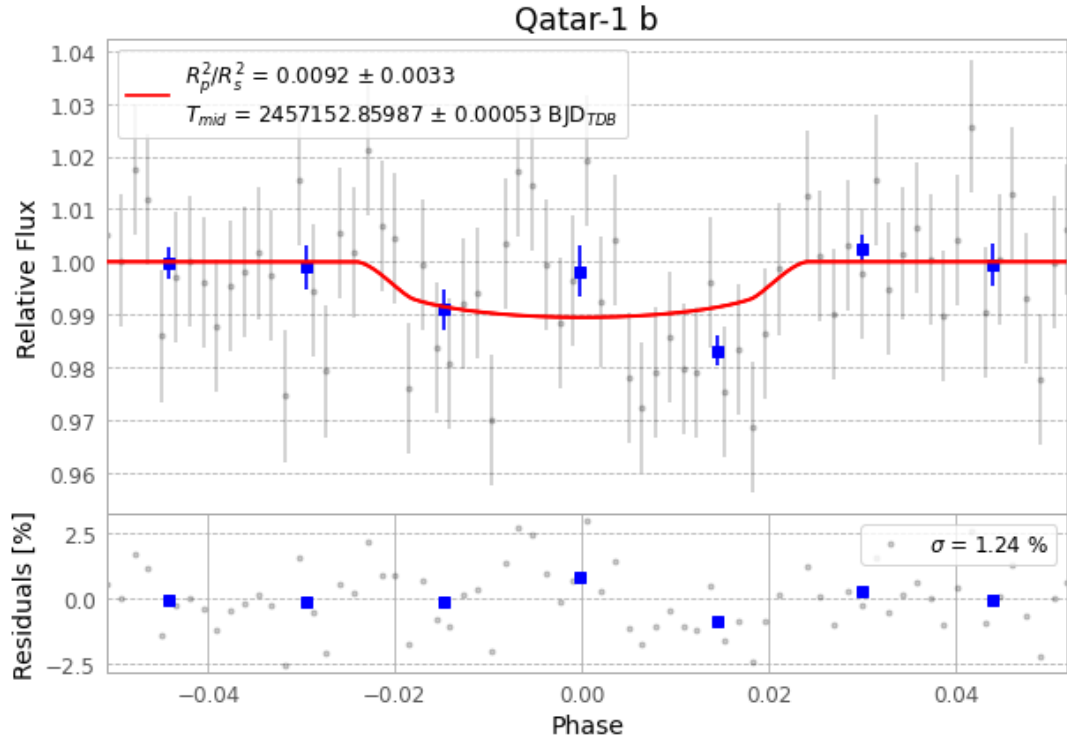


Figure 1 — FinalLightCurve_Qatar-1 b_2015-05-09.png (SHA-256 98b0eaa17e05badc...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [333, 296], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 6c21b90f8f75ff1e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

71 FITS frames (47 MB), Qatar-1150510065412.FITS → Qatar-1150510102412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-150510.log (da76df18a126...), FinalLightCurve_Qatar-1 b_2015-05-09.png (98b0eaa17e05...), FinalLightCurve_Qatar-1 b_2015-05-09.csv (b696a37f9400...)

Compute resources

Wall time 180.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 05:38Z.